

10.1 Warm-Up Exercises

10.1. Check the calculations in Examples 10.2, 10.3 and 10.4.

10.2. For each of the following matrices A , write down a quadratic equation with associated matrix A and find the matrix $M \in \text{SO}(2)$ which diagonalizes A .

a) $\begin{bmatrix} 6 & 2 \\ 2 & 9 \end{bmatrix}$

b) $\begin{bmatrix} 5 & -15 \\ -15 & 5 \end{bmatrix}$

c) $\begin{bmatrix} 7 & -2 \\ -2 & 5/3 \end{bmatrix}$

10.3. For the curve with equation $x^2 - 4xy + y^2 - 6x + 2y + 1 = 0$

- a) find the isometric canonical equation;
- b) find the affine canonical equation using Lagrange's method and indicate the corresponding affine change of coordinates.

10.2. For each of the following matrices A , write down a quadratic equation with associated matrix A and find the matrix $M \in SO(2)$ which diagonalizes A .

a) $\begin{bmatrix} 6 & 2 \\ 2 & 9 \end{bmatrix}$

- a quadratic equation with associated matrix is $6x^2 + 9y^2 + 4xy + 42 = 0$
- the matrix T is the base change matrix $M_{e',e}$ where e is the basis with respect to which the matrix A is given and e' is an orthonormal basis of eigenvectors

$$\det(\lambda I_2 - A) = \begin{vmatrix} \lambda - 6 & -2 \\ -2 & \lambda - 9 \end{vmatrix} = (\lambda - 6)(\lambda - 9) - 4 = \lambda^2 - 15\lambda + 54 - 4 = 0$$

$$\lambda^2 - 15\lambda + 50 = 0 \quad \Leftrightarrow \quad (\lambda - 10)(\lambda - 5) = 0$$

so, the eigenvalues are 5 and 10

$$\lambda = 5 \quad A \cdot \begin{bmatrix} x \\ y \end{bmatrix} = 5 \begin{bmatrix} x \\ y \end{bmatrix} \quad \Leftrightarrow \quad (5I - A) \begin{bmatrix} x \\ y \end{bmatrix} = 0$$

$$\Leftrightarrow \begin{bmatrix} -1 & 2 \\ 2 & -4 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = 0$$

$$\begin{bmatrix} -1 & 2 \\ 2 & -4 \end{bmatrix} \sim \begin{bmatrix} -1 & 2 \\ 0 & 0 \end{bmatrix} \quad \text{so } -x + 2y = 0 \\ \Rightarrow x = 2y$$

the eigenspace for the eigenvalue $\lambda = 5$ is

$$\left\{ \begin{bmatrix} 2t \\ t \end{bmatrix} : t \in \mathbb{R} \right\} = \left\langle \begin{bmatrix} 2 \\ 1 \end{bmatrix} \right\rangle$$

$$\lambda = 10 \quad (10I - A) \begin{bmatrix} x \\ y \end{bmatrix} = 0 \quad \Leftrightarrow \quad \begin{bmatrix} 4 & -2 \\ -2 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = 0$$

$$\begin{bmatrix} 4 & -2 \\ -2 & 1 \end{bmatrix} \sim \begin{bmatrix} 2 & -1 \\ 0 & 0 \end{bmatrix} \quad \text{so } 2x - y = 0 \Rightarrow y = 2x$$

the eigenspace for the eigenvalue $\lambda = 10$ is

$$\left\{ \begin{bmatrix} t \\ 2t \end{bmatrix} : t \in \mathbb{R} \right\} = \left\langle \begin{bmatrix} 1 \\ 2 \end{bmatrix} \right\rangle$$

$\Rightarrow$ an orthonormal basis of eigenvectors is $\left\{ \frac{1}{\sqrt{5}} \begin{bmatrix} -2 \\ 1 \end{bmatrix}, \frac{1}{\sqrt{5}} \begin{bmatrix} 1 \\ 2 \end{bmatrix} \right\}$

so M could be $M_{e_i e_i} = \frac{1}{\sqrt{5}} \begin{bmatrix} -2 & 1 \\ 1 & 2 \end{bmatrix}$ $M \cdot M^t (= M^2) = \frac{1}{5} \begin{bmatrix} 5 & 0 \\ 0 & 5 \end{bmatrix} = I_2$

$$\det M = \frac{1}{5} \cdot (-5) = -1$$

$\Downarrow$

we can replace $\frac{1}{\sqrt{5}} \begin{bmatrix} -2 \\ 1 \end{bmatrix}$ by $\frac{1}{\sqrt{5}} \begin{bmatrix} 2 \\ -1 \end{bmatrix}$

then $M_{e_i e_i} = \frac{1}{\sqrt{5}} \begin{bmatrix} 2 & 1 \\ -1 & 2 \end{bmatrix}$ has $M \cdot M^t = \frac{1}{5} \begin{bmatrix} 2 & 1 \\ -1 & 2 \end{bmatrix} \cdot \begin{bmatrix} 2 & -1 \\ 1 & 2 \end{bmatrix} = \frac{1}{5} \begin{bmatrix} 5 & 0 \\ 0 & 5 \end{bmatrix} = I_2$

and $\det M = \frac{1}{5} \cdot 5 = 1$

$$\Rightarrow M \in SO(2)$$

Moreover

$$M^{-1} A M = M^t A M = \frac{1}{\sqrt{5}} \begin{bmatrix} 2 & -1 \\ 1 & 2 \end{bmatrix} \cdot \begin{bmatrix} 6 & 2 \\ 2 & 9 \end{bmatrix} \cdot \frac{1}{\sqrt{5}} \begin{bmatrix} 2 & 1 \\ -1 & 2 \end{bmatrix}$$

$$= \frac{1}{5} \begin{bmatrix} 2 & -1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 10 & 10 \\ -5 & 20 \end{bmatrix} = \frac{1}{5} \begin{bmatrix} 25 & 0 \\ 0 & 50 \end{bmatrix} = \begin{bmatrix} 5 & 0 \\ 0 & 10 \end{bmatrix}$$

10.3. For the curve with equation $x^2 - 4xy + y^2 - 6x + 2y + 1 = 0$

a) find the isometric canonical equation;

associated matrix is $Q = \begin{bmatrix} 1 & -2 \\ -2 & 1 \end{bmatrix}$

eigenvalues and eigenvectors are

$$\lambda_1 = 3 \quad v_1 = (-1, 1)$$

$$\lambda_2 = -1 \quad v_2 = (1, 1)$$

$\Rightarrow$ orthonormal basis of eigenvectors is

$$B' = \left(\frac{1}{\sqrt{2}} \begin{bmatrix} -1 \\ 1 \end{bmatrix}, \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix} \right)$$

base change matrix

$$M = M_{BB'} = \frac{1}{\sqrt{2}} \begin{bmatrix} -1 & 1 \\ 1 & 1 \end{bmatrix} \text{ with determinant } -1$$

we change the sign of first vector

$$B' = \left(\frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \end{bmatrix}, \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix} \right)$$

then $M = M_{BB'} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}$ is a rotation by $\theta = -45^\circ$

the equation of our quadratic curve changes as follows

$$(x' y') M^T Q M \begin{bmatrix} x' \\ y' \end{bmatrix} + [-6 \ 2] M \begin{bmatrix} x' \\ y' \end{bmatrix} + 1 = 0$$

$$(\Rightarrow) (x' y') \underbrace{\frac{1}{\sqrt{2}} \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & -2 \\ -2 & 1 \end{bmatrix} \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}}_{\begin{bmatrix} 3 & 0 \\ 0 & -1 \end{bmatrix}} \begin{bmatrix} x' \\ y' \end{bmatrix} + [-6 \ 2] \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} x' \\ y' \end{bmatrix} + 1 = 0$$

$$(x' y') \begin{bmatrix} 3 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} x' \\ y' \end{bmatrix} + \frac{1}{\sqrt{2}} \begin{bmatrix} -8 & -4 \end{bmatrix} \begin{bmatrix} x' \\ y' \end{bmatrix} + 1 = 0$$

$$\Leftrightarrow x_2^2 - y_2^2 = -\frac{1}{3} \quad | : -\frac{1}{3}$$

$$\begin{pmatrix} x_3 \\ y_3 \end{pmatrix} = \begin{pmatrix} 0 & \sqrt{3} \\ \sqrt{3} & 0 \end{pmatrix} \begin{pmatrix} x_2 \\ y_2 \end{pmatrix}$$

$$\Leftrightarrow \underbrace{\left(\frac{y_2}{\frac{1}{\sqrt{3}}}\right)^2}_{x_3} - \underbrace{\left(\frac{x_2}{\frac{1}{\sqrt{3}}}\right)^2}_{y_3} = 1$$

$$\Leftrightarrow x_3^2 - y_3^2 = 1 \quad \text{hyperbola}$$

$$\text{So } \begin{pmatrix} x_3 \\ y_3 \end{pmatrix} = \begin{pmatrix} 0 & \sqrt{3} \\ \sqrt{3} & 0 \end{pmatrix} \left(\begin{pmatrix} 1 & 0 \\ 0 & \sqrt{3} \end{pmatrix} \begin{pmatrix} 1 & -2 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} -3 \\ \frac{5}{\sqrt{3}} \end{pmatrix} \right)$$

$$= \sqrt{3} \cdot \begin{pmatrix} 0 & \sqrt{3} \\ 1 & -2 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} 5 \\ -\sqrt{3} \end{pmatrix}$$

(the affine change of coordinates)